

Macroeconomic Agent-Based Models

Steven Hoekstra ^{1 2}

¹CeNDEF, University of Amsterdam,

²Tinbergen Institute



UNIVERSITY OF AMSTERDAM
Amsterdam School of Economics

Lecture overview

- 1 Introduction
- 2 Poledna et al. model
- 3 Calibration
- 4 External Validation
- 5 Advanced Calibration Methods
- 6 Application: CANVAS wage-price spiral
- 7 Discussion

From Qualitative to Quantitative ABMs

- Recall the Assenza et al. (2015) model from last lecture: a qualitative macro ABM that generates emergent business cycles.
- Today we move to **quantitative** ABMs: the Poledna et al. (2023) model calibrated to real data.
- We then discuss advanced calibration techniques to identify the remaining free parameters.

Model overview

- All economic activities classified by the **European System of Accounts** (ESA 2010).
- **Complete GDP identity** via the production, income, and expenditure approaches simultaneously.
- **Adaptive learning** instead of rational expectations.
- **Decentralized markets** via search-and-matching with **trade frictions**.
- **Multi-sector production network** calibrated from input-output tables.
- **Stock-flow consistent** accounting across all sectors.
- **Non-linear amplification** of shocks generates **endogenous crises** without requiring large exogenous disturbances.

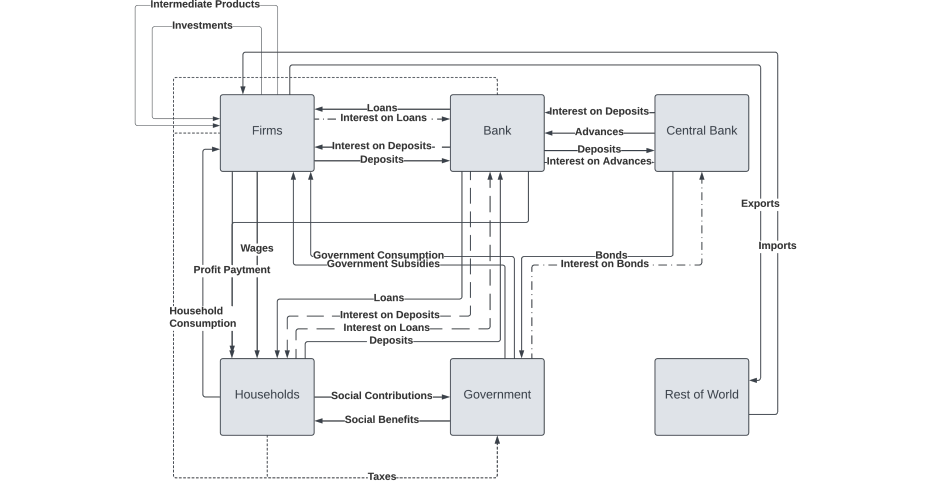


Figure 1. Flow diagram of the model

Data-based initialisation & no burn-in

Why the Poledna model needs no burn-in:

1. **SFC accounting matches ESA 2010.** The model's sectoral accounting *is* the national accounts framework. At $t=0$, balance sheets and flows are loaded directly from Austrian Eurostat data.
2. **The initialisation is a quasi-fixed point..** If we suppress the growth processes, the model fluctuates around it's initial conditions.
3. **Contrast with standard ABMs.** Most agent-based models start from an arbitrary or stylised initial state that is *not* a rest point of the dynamics. Even without shocks, the model must first converge to an attractor.

Caveat: the AR(1) expectation parameters and five exogenous shock processes are still estimated on pre-sample data (1997:Q1–). There is no dynamic burn-in, but there is a historical estimation window.

Sequence of events in each quarter

Each period t is one quarter. The model is solved recursively:

- (i) **Expectations:** agents OLS-update their AR(1) rules for output and inflation using last period's realisation.
- (ii) **Exogenous shocks:** draw from the AR(1) processes for exports, imports, government consumption, euro-area growth, euro-area inflation.
- (iii) **Credit and labour markets:** search-and-matching solved. Firms apply for loans and hire/fire; unemployed households seek jobs.
- (iv) **Goods market:** search-and-matching solved. Households, firms (intermediate inputs), government, and foreign consumers meet firms; production and sales happen.
- (v) **Accounting:** update all stock and flow variables; check insolvencies; compute profits, taxes, transfers.

Each reference quarter is simulated **500 times** (Monte Carlo); the forecast is the average across runs.

Household behaviour: consumption, savings, expectations

Consumption, Savings & Expectations:

- Expected income depends on activity status and expectations.
- Consumption budget: $C_h^d(t) = \frac{\psi Y_h^e(t)}{1+\tau^{VAT}}$ where $\psi \in (0, 1)$ is propensity to consume
- Budget allocation: $C_{hg}^d(t) = \frac{b_g^{HH} P_g(t-1)}{P^{HH}(t-1)} C_h^d(t)$ where b_g^{HH} is consumption coefficient
- $D_h(t) = D_h(t-1) + Y_h(t) - (1 + \tau^{VAT})C_h(t) - (1 + \tau^{CF})I_h(t) + \text{interest}$
- Expectations: agents use optimal AR(1) forecasting rules updated each period via OLS,

$$\mathbb{E}_t^*[\pi_{t+1}] = \alpha_t^\pi \pi_t + \beta_t^\pi, \quad \mathbb{E}_t^*[y_{t+1}] = \alpha_t^y y_t + \beta_t^y.$$

Household heterogeneity: four types

- **Employed** (H_t^E): supply labour to a specific sector, earn sector-specific wage $w_{i,t}$.
- **Unemployed** (H_t^U): involuntarily idle, receive benefits $\theta^{UB} w_{h,t}$.
- **Investor**: one per firm, owns firm equity, receives dividends $\theta^{DIV} (1 - \tau^{FIRM}) \Pi_{i,t}$.
- **Inactive** (H^{inact}): outside the labour force, receive social transfers sb_t^{inact} .

Common to all households:

- same MPC ψ
- invest fraction ψ^H of income in dwellings
- purchase consumption via goods-market S&M

Heterogeneity is what lets the model track **distributional effects** through the income and capital accounts.

Firms

- Firms produce output using labour, intermediate inputs and raw material, and capital using a **Leontief** production function

$$Y_{i,t} = \min \left(Q_{i,t}^S, \beta_i M_{i,t}, \alpha_{i,t} N_{i,t}, \kappa_i K_{i,t-1} \right).$$
- Firms face demand via the **decentralized markets**.
- Firms set quantity and price based on **boundedly rational expectations** of **growth and inflation**.
- Firms can finance expenditures by **borrowing** from the bank.
- Firms can become **bankrupt** if they are both insolvent and have negative equity value.

Labour market: search and matching

- The labour market is solved **before** the goods market each quarter.
- Firms post vacancies consistent with their expected production and the Leontief labour coefficient: $N_{i,t}^d = Q_{i,t}^s / \alpha_{i,t}$.
- Unemployed households apply to firms in random order; matching probability is weighted by **firm size** and **posted wage**.
- Hiring/firing is frictional: if $N_{i,t}^d < N_{i,t-1}$ firms lay off the excess; if $N_{i,t}^d > N_{i,t-1}$ they hire from the unemployed pool.
- Wages update slowly around the sector-specific average $\bar{w}_i$, indexed to expected inflation.

Input-output network

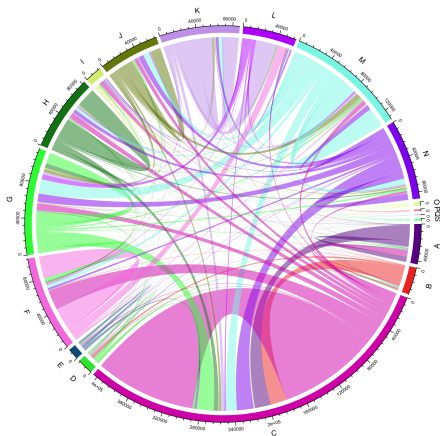


Figure: Intermediate consumption between 19 industries derived from Eurostat input-output table.

Remainder of model

- **Government** consumes on the goods market, levies taxes, pays out transfers and social benefits.
- **Banking sector** provides loans to firms subject to a minimum capital requirement and a risk assessment; holds household and firm deposits.
- **Central bank** sets the policy rate via a Taylor rule and holds government debt as an asset.
- **Rest of the world** exports to and imports from the domestic economy via the decentralized markets.

Fiscal and monetary closure

Government (fiscal)

- **Revenue:** income tax, VAT, corporate tax, social contributions (employer, employee), capital-formation tax, export tax, sector-specific production/product taxes.
- **Spending:** government consumption, unemployment benefits at replacement rate, inactive transfers, interest on debt.
- **Debt dynamics:**

$$L_t^G = L_{t-1}^G + (\text{expenditures}_t - \text{revenues}_t)$$

Every tax, transfer, loan, and interest payment is recorded inside the SFC structure: the government and banking sectors are *part of* the balance-sheet closure, not black boxes bolted on.

Bank & central bank (monetary)

- **Basel III capital requirement:**
 $E_{k,t} / \sum_i L_{i,t} \geq \zeta = 3\%$. When bad-debt write-offs shrink bank equity $E_{k,t}$, the constraint binds and credit to *other* firms is rationed, this is the credit channel of the bankruptcy cascade.
- **Loan-to-value** on new loans:
 $L_i / (\bar{P}^{\text{CF}} K_i) \leq \zeta^{\text{LTV}} = 60\%$.
- **Loan rate:** $r_t = \bar{r}_t + \mu$, fixed markup $\mu = 2.9\%$ over the policy rate.
- **Policy rate $\bar{r}_t$:** generalised Taylor rule reacting to *euro-area* inflation and growth (next slide).

What drives fluctuations? Five exogenous AR(1) processes

- Unlike Assenza et al. (2015), where fluctuations are entirely endogenous, the Poledna ABM is driven by **five exogenous AR(1) processes** estimated from data:
 - 1 Export demand C_t^E
 - 2 Import supply Y_t^I
 - 3 Government consumption C_t^G
 - 4 Euro-area GDP growth γ_t^{EA}
 - 5 Euro-area inflation π_t^{EA}
- The shocks are correlated via a covariance matrix **C** estimated from national accounts.
- Euro-area growth and inflation feed directly into the Taylor rule $\Rightarrow$ monetary policy reacts to the euro-area cycle, not just Austrian conditions (small open economy).
- Endogenous fluctuations are *amplified* by these shocks but not *caused* by them; bankruptcy cascades generate their own crises (see next).

Endogenous crises: the bankruptcy cascade

- A firm becomes insolvent when **both**: cash-flow negative ($D_{i,t} < 0$) and balance-sheet negative ($E_{i,t} < 0$).
- Insolvency triggers a **cascade** through three channels:
 - **Labour**: workers at the failing firm are laid off → household income drops → consumption falls.
 - **Supply chain**: upstream suppliers lose a buyer → inventory build-up → their profits fall.
 - **Credit**: the bank writes off bad debt → $E_{k,t}$ drops → the minimum capital requirement binds → credit to *other* firms is rationed.
- Each channel is non-linear and **path-dependent**: the economy does not return to the BLE it came from.
- Multiple bankruptcies amplify the initial shock into a **medium-term recession** without any further exogenous disturbance.
- This is the mechanism behind “endogenous crises”, the feature that distinguishes the ABM from linearised DSGE, where balance-sheet constraints do not bind.

Data Sources

- CBS Register data and Eurostat data
- Census and business demography data
- Input-output tables
- Government statistics and sector accounts
- Statutory guidelines, financial regulation (Basel III), and banking practices

How big is the model?

Agent counts (Austria, 2010:Q4)

Economically active persons	4,729,215
Inactive persons	4,130,385
Firms / investors ¹	≈ 611,280
Government entities	152,820
Foreign consumers	305,639
Foreign firms (imports)	≈ 50

Total: ~ 9.9 million heterogeneous agents.

Derived from the paper's Table 2: $J = 25\%$ of domestic firms, $L = 50\%$; both give $\approx 611,280$.

Sector structure

- **64 NACE industries** (the I-O figure earlier showed an aggregated 19 for visual clarity)
- Firm size distribution within each industry follows a **power law**, calibrated from Eurostat business demography
- Industry-specific: productivities $\bar{\alpha}_i, \kappa_i, \beta_i$, depreciation δ_i , wages $\bar{w}_i$
- No representative firm, no representative consumer

Calibration Process

- The biggest innovation of the model is that almost all parameters are directly identified from Eurostat national accounts.
- At time $t = 0$, the initial values of the model match the data found in the national accounts (or are imputed from the same data at the micro level).
- Combined with stock-flow consistency, this means no dynamic burn-in is required (see earlier slide).
- The parameters are identified using the datasets listed on the previous slide.
- For example, the marginal propensity to consume ψ is calibrated as the ratio of household consumption to disposable income.
- A small number of parameters still may be inferred. More on this later.

The full parameter set: (almost) no free parameters

Source	Example parameters	Value
Census / business demography	Active persons H^{act} Firms per industry I_s	4,729,215 power law
Input-output tables	Technology coefs a_{sg} Labour productivity $\bar{\alpha}_i$ Consumption coefs b_g^{HH}	matrix industry by product
Gov. statistics / sector accounts	Income tax τ^{INC} VAT τ^{VAT} Corporate tax τ^{FIRM} MPC ψ Dividend payout θ^{DIV}	0.2134 0.1529 0.0762 0.9394 0.7768
Basel III / statutes	Capital ratio ζ LTV ζ^{LTV} Unempl. benefit θ^{UB} Inflation target π^*	0.03 0.60 0.3586 0.005
AR(1) & Taylor rule (estimated)	Taylor smoothing ρ Inflation weight ξ^π Growth weight ξ^γ	0.9263 0.3214 1.2994

~ 50 parameters total. Only the AR(1) / Taylor-rule block is estimated; everything else is read off data or statute.

Out-of-sample forecast performance in comparison to DSGE model

	GDP	Inflation	Household consumption	Investment	Euribor
AR(1)	<i>RMSE-statistic for different forecast horizons</i>				
1q	0.52	0.3	0.52	1.18	0.03
2q	0.77	0.28	0.73	1.78	0.06
4q	1.26	0.28	1.14	2.91	0.11
8q	2.12	0.29	1.93	4.26	0.16
12q	2.89	0.25	2.74	6.01	0.2
DSGE	<i>Percentage improvements (+) or losses (−) relative to AR(1) model</i>				
1q	1.1 (0.93)	−13.6 (0.05**)	7.7 (0.25)	−1.6 (0.87)	−147.8 (0.00***)
2q	−5.7 (0.70)	−10.9 (0.19)	12.5 (0.20)	2.4 (0.90)	−183.4 (0.00***)
4q	−8.2 (0.76)	−9.6 (0.09*)	30.5 (0.21)	−10.5 (0.74)	−201.3 (0.00***)
8q	−2 (0.97)	2.7 (0.80)	38.5 (0.29)	−19.4 (0.57)	−210 (0.00***)
12q	3.7 (0.97)	−14.9 (0.18)	49 (0.30)	0.4 (0.98)	−180.4 (0.00***)
ABM	<i>Percentage improvements (+) or losses (−) relative to AR(1) model</i>				
1q	1.7 (0.88)	−0.2 (0.80)	−16.1 (0.26)	−3.7 (0.12)	−3.6 (0.44)
2q	−1.8 (0.90)	−0.7 (0.52)	−13.1 (0.56)	−4.3 (0.23)	−2.6 (0.69)
4q	−5.5 (0.81)	1.5 (0.03*)	−2 (0.95)	−4.6 (0.26)	2.1 (0.85)
8q	−3.4 (0.97)	1.2 (0.53)	11.8 (0.83)	−6.9 (0.56)	0.6 (0.98)
12q	−2.7 (0.97)	0.2 (0.80)	19.7 (0.79)	−5.7 (0.71)	−0.3 (0.99)

Note: Forecast period 2010:Q2–2019:Q4. The AR(1) and the DSGE model are re-estimated each quarter. The ABM is calibrated to 39 different reference quarters. ABM results are averaged over 500 Monte Carlo simulations. *p*-values in parentheses are from (modified) Diebold–Mariano tests (Harvey et al., 1997), under the null that the tested model has the same accuracy as the AR(1). *, **, *** denote significance at the 10, 5, and 1 per cent levels, respectively. Source: Poledna et al. (2023), Table 4.

Out-of-sample forecast performance

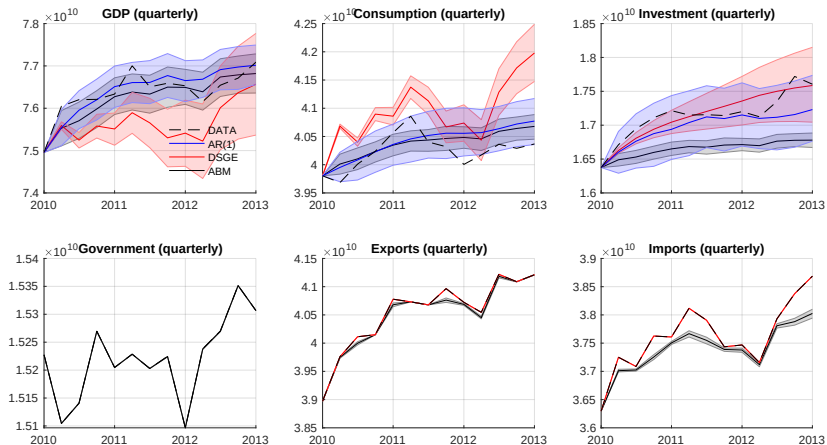


Figure: Forecasting performance for one reference quarter (2012:Q4–2014:Q4): ABM conditional (black), DSGE conditional (red), ARX(1) (blue), and observed Eurostat data for Austria (dashed). Previous slide shows full-period RMSE (2010:Q1–2019:Q4).

Quarterly out-of-sample GDP forecast decomposition

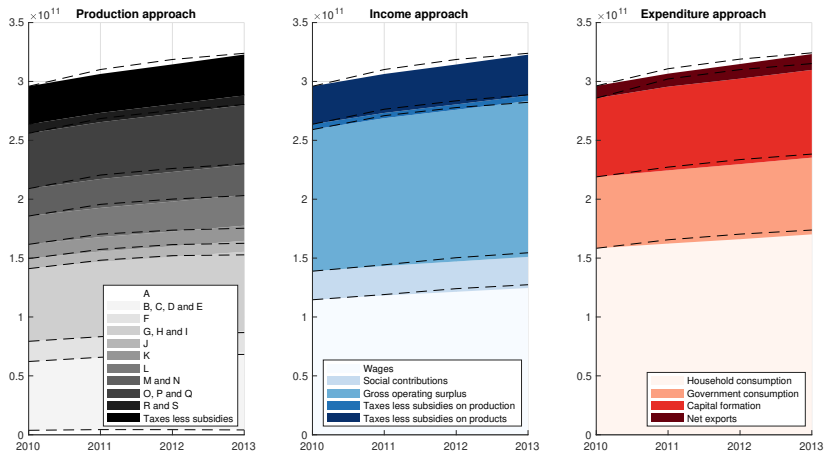


Figure: The full coloured areas are the average ABM forecast results.

Poledna et al. (2023) summary

- The Poledna ABM performs **as well as or better than** the comparison models, depending on variable and horizon (not all results shown).
- The model is validated primarily via **out-of-sample forecasting** against observed data, in contrast to Assenza et al. (2015), which was only validated against stylised facts.
- Performance improves further in a **conditional forecasting** setup where exports, imports, and government consumption are fixed at realised paths.
- For sectoral GVA forecasts the ABM beats the VAR in some sectors and underperforms in others (e.g. agriculture, real estate).
- A great starting point for further extensions.
- These models are gaining traction academically and in the policy world, e.g. the ECB, the Bank of England, the Bank of Canada, and the Bank of Italy.

The Likelihood Problem in Agent-Based Models

Goal: Bayesian inference for ABM parameters

$$p(\theta|x_{\text{obs}}) = \frac{p(x_{\text{obs}}|\theta)p(\theta)}{p(x_{\text{obs}})}$$

The Problem: Likelihood $p(x|\theta)$ is intractable. We can simulate $x \sim \text{ABM}(\theta)$, but cannot evaluate $p(x|\theta)$

Why standard Bayesian methods fail:

- *MCMC*: Acceptance ratio $\alpha = \min \left(1, \frac{p(x|\theta')p(\theta)}{p(x|\theta)p(\theta')} \cdot \frac{q(\theta|\theta')}{q(\theta'|\theta)} \right)$ requires likelihood evaluation at every step
- *Variational Inference*: Evidence Lower Bound (ELBO)
 $= \mathbb{E}_{q_\phi} [\log p(x|\theta)] - D_{KL}(q_\phi \| p(\theta))$ requires likelihood in expectation
- *Approximate Bayesian Computation*: Works, but acceptance rate suffers from the curse of dimensionality.

Neural Posterior Estimation (Dyer et al., 2024)

- **Problem:** Need posterior $p(\theta|x)$ but the likelihood $p(x|\theta)$ is intractable for ABMs.
- **Solution: Neural Posterior Estimation:**
 - Generate training data $\{x_i, \theta_i\}_{i=1}^{N_{\text{sims}}}$ by sampling $\theta_i \sim \Phi$ (prior) and simulating $x_i \sim \text{ABM}(\theta_i)$.
 - Train neural network $q_\phi(\theta|x)$ to approximate posterior:

$$\hat{\phi}^{(r)} = \arg \max_{\phi} \sum_{i=1}^{N_{\text{sims}}} \log q_{\phi}(\theta_i | x_i)$$

- Refine with sequential rounds using current posterior as new prior:

$$\theta_i^{(r+1)} \sim q_{\hat{\phi}(r)}(\theta | x_{\text{obs}})$$

- For any observed data x_{obs} , directly sample from final posterior:

$$\theta \sim q_{\hat{\phi}^{\text{final}}}(\theta|x_{\text{obs}})$$

- **Key advantage:** Overcomes intractable likelihood by learning the posterior directly, bypassing likelihood evaluation entirely and enabling full posterior uncertainty quantification.

Sequential Neural Posterior Estimation (SNPE)

Motivation: a single round of NPE wastes simulations on regions of Θ that turn out to have low posterior density given the observed $\mathbf{y}$. SNPE narrows the search iteratively by re-using the current posterior estimate as the proposal distribution for the next batch of simulations, concentrating compute where it matters.

Algorithm Sequential Neural Posterior Estimation (SNPE)

- 1: Initialise proposal $\tilde{p}_0(\theta) = p(\theta)$; dataset $\mathcal{D} = \emptyset$
- 2: **for** $m = 0, \dots, M - 1$ rounds **do**
- 3: Sample $\theta^{(n)} \sim \tilde{p}_m(\theta)$; simulate $\mathbf{x}^{(n)} \sim p(\mathbf{x}|\theta^{(n)})$; append to $\mathcal{D}$
- 4: Train the conditional flow $q_\phi(\theta|\mathbf{x})$ with the **importance-reweighted** loss

$$\mathcal{L}(\phi) = - \sum_{(\mathbf{x}, \theta) \in \mathcal{D}} \log \left(q_\phi(\theta|\mathbf{x}) \cdot \frac{\tilde{p}_m(\theta)}{p(\theta)} \cdot \frac{1}{Z(\mathbf{x}, \phi)} \right)$$

- 5: Update proposal: $\tilde{p}_{m+1}(\theta) := q_\phi(\mathbf{y}, \theta)$
 - 6: **end for**
-

The importance weight $\tilde{p}_m/p$ is essential: after round 1, parameters are no longer drawn from the prior $p(\theta)$ but from the (narrower) proposal $\tilde{p}_m(\theta)$. The reweighting corrects for this shift; without it, SNPE converges to the wrong distribution.

Results: decisive improvement over KDE

Benchmark 1: Brock and Hommes (1998): four heuristic belief types with intensity of choice β , parameter $\theta = (g_2, b_2, g_3, b_3)$.

	KDE	SNPE
Wasserstein (param set 1)	0.690	0.336
MMD	1.015	0.552
Simulation budget	1.5×10^5	10^4

Benchmark 2: Multivariate Geometric Brownian Motion: $d = 3$ drift/volatility inference.

	KDE	NPE
Wasserstein	0.364	0.099
MMD	0.137	0.005
Simulation budget	10^6	10^3

10–1000 $\times$ reduction in simulation budget, and better posterior recovery.

Visual results: Brock–Hommes posteriors

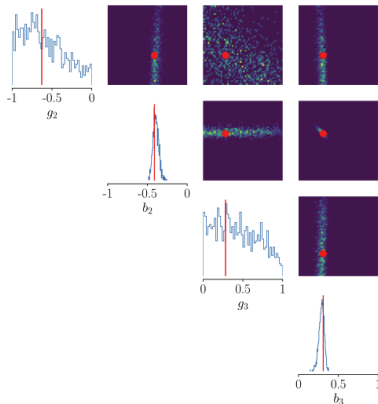


Figure: SNPE with learned summary statistics on the Brock and Hommes (1998) asset-pricing model. Marginal posteriors on the diagonal, bivariate joints above; red lines show the true posterior mean. Source: Dyer et al. (2024), Fig. 3d. Code: joelnmdyer/sbi4abm (a wrapper around the Mackelab sbi package).

Pricing mechanism

- The CANVAS model uses the **expected marginal cost + markup rule**.

$$P_i(t) = MC_i^e(t) \cdot \mu_i(t)$$

- Marginal cost is determined by the sum of all production costs.

$$MC_i(t) = \underbrace{\frac{(1 + \tau^{\text{SIF}}(t)) \bar{w}_i(t)}{\bar{\alpha}_i(t)} P^w(t)}_{\text{unit labour costs}} + \underbrace{\frac{1}{\beta_i(t)} \sum_g a_{sg}(t) \bar{P}_g(t)}_{\text{unit intermediate costs}} + \underbrace{\frac{\delta_i(t)}{\kappa_i(t)} \bar{P}^{CF}(t)}_{\text{unit capital costs}}.$$

- Marginal costs are inflated by expected inflation.

$$MC_i^e(t) = (1 + \pi^e(t)) \cdot MC_i(t-1).$$

- The markup is updated by demand pull (explained next slide).

$$\mu_i(t) = (1 + \pi^{DP}(t)) \mu_i(t-1)$$

Demand pull mechanisms

- Firms form firm-level expectations using two indicators from the previous period: **excess supply** (difference between supply and realized demand) and their **price deviation** from competitors' average price.
- Based on these conditions, firms follow four strategic scenarios that determine whether to adjust price or production:
 - clear inventory by decreasing prices,
 - decrease production to match demand,
 - capture profitable market opportunity, and
 - increase prices to increase margins.

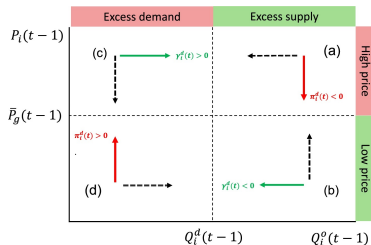


Figure: Source: Hommes et al. (2025)

Demand pull mechanisms (formulae)

- Demand pull inflation is determined by (scenario (a) and (d))

$$\pi_i^d(t) = \begin{cases} \frac{Q_i^d(t-1)}{Q_i^o(t-1)} - 1 & \text{if } Q_i^o(t-1) \leq Q_i^d(t-1) \text{ and } P_i(t-1) < \bar{P}_g(t-1) \\ & \text{or if } Q_i^o(t-1) > Q_i^d(t-1) \text{ and } P_i(t-1) \geq \bar{P}_g(t-1) \\ 0 & \text{otherwise.} \end{cases}$$

- Quantity setting is given by $Q_i^s(t) = Q_i^o(t-1)(1 + \gamma^e(t))(1 + \gamma_i^d(t))$, where $\gamma^e(t)$ is the aggregate growth rate and $\gamma_i^d(t)$ is idiosyncratic growth.
- $\gamma_i^d(t)$ is determined by (scenario (b) and (c))

$$\gamma_i^d(t) = \begin{cases} \frac{Q_i^d(t-1)}{Q_i^o(t-1)} - 1 & \text{if } Q_i^o(t-1) \leq Q_i^d(t-1) \text{ and } P_i(t-1) \geq \bar{P}_g(t-1) \\ & \text{or if } Q_i^o(t-1) > Q_i^d(t-1) \text{ and } P_i(t-1) < \bar{P}_g(t-1) \\ 0 & \text{otherwise.} \end{cases}$$

CANVAS pricing mechanism implicit assumptions

- The CANVAS pricing equation can be written as

$$P_i(t) = P_i(t-1) \cdot \underbrace{\left(1 + \phi^{DP} \pi_i^d(t)\right)}_{\text{Demand-pull inflation}} \cdot \underbrace{\left(1 + \phi^{CP} \pi_i^c(t)\right)}_{\text{Cost-push inflation}} \cdot \underbrace{\left(1 + \phi^{AE} \pi^e(t)\right)}_{\text{Aggregate inflation expectation}}$$

- However, the CANVAS model implicitly assumes that $\phi^{DP} = \phi^{CP} = \phi^{AE} = 1$.
- This assumes all inflation costs pass through to prices, but firms actually absorb some costs.
- Moreover, it assumes that wages are indexed fully and that the quantity demand-pull channel has 100% strength.

Sectoral inflation decomposition: wage shock, non-estimated ABM

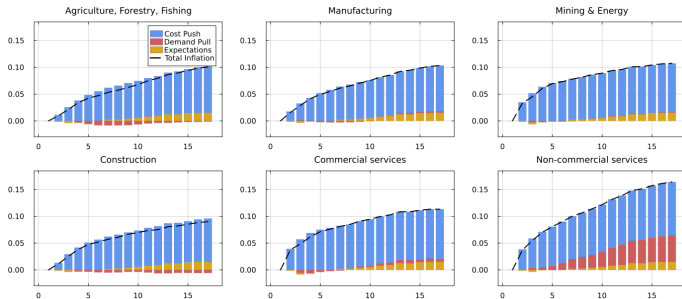


Figure: Sectoral inflation decomposition following 10% wage shock. Stacked bars show contributions from cost pass-through, demand pull, and expectations components. Dashed line indicates total cumulative inflation difference from baseline simulation over 16 quarters.

Sectoral inflation decomposition: wage shock, estimated ABM

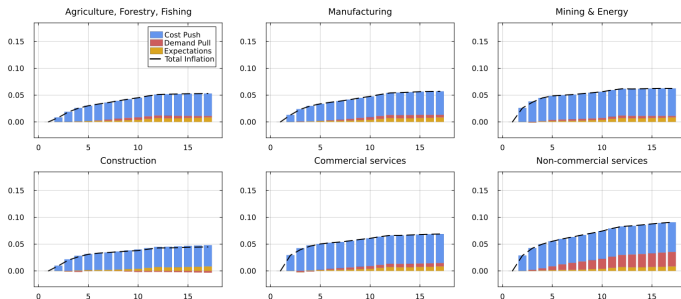


Figure: Sectoral inflation decomposition following 10% wage shock. Stacked bars show contributions from cost push, demand pull, and expectations inflation. Dashed line indicates total cumulative inflation difference from baseline simulation over 16 quarters.

Further sectoral inflation decomposition: non-estimated ABM

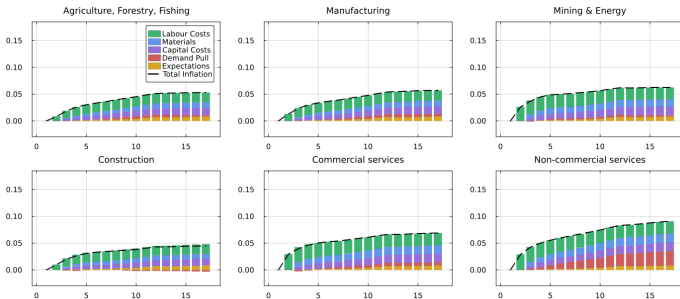


Figure: Sectoral inflation decomposition following 10% wage shock. Stacked bars show contributions from demand pull inflation, labour costs, intermediate products, capital costs, and aggregate inflation. Dashed line indicates total cumulative inflation difference from baseline simulation over 16 quarters.

Concluding Remarks

- We have discussed the **qualitative** Assenza et al. (2015) model in detail.
- We have covered the **quantitative** agent-based model of Poledna et al. (2023) and its calibration to Eurostat national-accounts data.
- We have covered **simulation-based Bayesian inference** for ABMs via neural posterior estimation (Dyer et al., 2024).
- We are now at the frontier of agent-based modelling!

Thank you very much for your attention!

Questions?

Bibliography I

- T. Assenza, D. Delli Gatti, and J. Grazzini. Emergent dynamics of a macroeconomic agent based model with capital and credit. *Journal of Economic Dynamics and Control*, 50:5–28, Jan. 2015.
- T. S. Blattner and E. Margaritov. Towards a Robust Monetary Policy Rule for the Euro Area. *SSRN Electronic Journal*, 2010.
- W. A. Brock and C. H. Hommes. Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control*, 22(8):1235–1274, 1998.
- J. Dyer, P. Cannon, J. D. Farmer, and S. M. Schmon. Black-box Bayesian inference for agent-based models. *Journal of Economic Dynamics and Control*, 161:104827, Apr. 2024.
- A. G. Haldane and A. E. Turrell. An interdisciplinary model for macroeconomics. *Oxford Review of Economic Policy*, 34(1-2):219–251, Jan. 2018.

Bibliography II

- D. Harvey, S. Leybourne, and P. Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, June 1997.
- C. Hommes, M. He, S. Poledna, M. Siqueira, and Y. Zhang. CANVAS: A Canadian behavioral agent-based model for monetary policy. *Journal of Economic Dynamics and Control*, 172:104986, Mar. 2025.
- S. Poledna, M. G. Miess, C. Hommes, and K. Rabitsch. Economic forecasting with an agent-based model. *European Economic Review*, 151:104306, Jan. 2023.
- J. B. Taylor. Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39:195–214, Dec. 1993.

Decentralized market

- Each firm i experiences demand $Q_{i,t}^d$ from consumers. Realized sales depend on production and inventory.

$$Q_{i,t}^d \begin{cases} < Y_{i,t} + S_{i,t-1} & \text{if demand from consumers is smaller than supply from firm } i, \text{ and} \\ = Y_{i,t} + S_{i,t-1} & \text{if demand from consumers exactly matches supply from firm } i, \text{ and} \\ > Y_{i,t} + S_{i,t-1} & \text{if demand from consumers is larger than supply from firm } i, \end{cases} \quad (1)$$

- Sales $Q_{i,t}$ are the realized demand dependent on the supply available from firm i after the production process has taken place:

$$Q_{i,t} = \min \left(S_{i,t-1} + Y_{i,t}, Q_{i,t}^d \right)$$

- Matching probabilities depend on size of the firm and on the price

$$\begin{aligned} pr_{i,t}^{\text{price}} &= \frac{e^{-2P_{i,t}}}{\sum_{i \in I_{S=g}} e^{-2P_{i,t}}} \\ pr_{i,t}^{\text{size}} &= \frac{Y_{i,t}}{\sum_{i \in I_{S=g}} Y_{i,t}} \\ pr_{i,t}^{\text{cum}} &= \frac{pr_{i,t}^{\text{price}} + pr_{i,t}^{\text{size}}}{2} \end{aligned} \quad (2)$$

[Return](#)

Quantity and Price Setting

- Quantity setting is based on expected growth rates

$$Q_{i,t}^s = Q_{i,t-1}^d (1 + \gamma_t^e). \quad (3)$$

- Expectations are formed using AR(1) rule or VAR(1) rule
- Prices are determined the previous set price, expected inflation and the cost-structure of the firm:

$$P_{i,t} = P_{i,t-1} \cdot \underbrace{\left(1 + \pi_{i,t}^c\right)}_{\text{Cost-Push Inflation}} \cdot \underbrace{\left(1 + \pi_t^e\right)}_{\text{Built-In Inflation}},$$

where

$$\pi_{i,t}^c = \underbrace{\frac{(1 + \tau^{SIF}) \bar{w}_i}{\bar{\alpha}_i} \left(\frac{\bar{P}_t^{HH}}{P_{i,t}} - 1 \right)}_{\text{Increase of unit labour costs}} + \underbrace{\frac{1}{\beta_i} \left(\frac{\sum_g a_{sg} \bar{P}_{g,t-1}}{P_{i,t}} - 1 \right)}_{\text{Increase of unit material costs}} + \underbrace{\frac{\delta_i}{\kappa_i} \left(\frac{\bar{P}_{t-1}^{CF}}{P_{i,t}} - 1 \right)}_{\text{Increase of unit capital costs}} \quad \forall i \in I_S. \quad (4)$$

Return

External Finance

It is possible that firms need to finance current or future expenditures by borrowing. Therefore, each firm forecasts future cash flow $\Delta D_{i,t}^e$, which is equal to the expected change in deposits $D_{i,t}$:

$$\Delta D_{i,t}^e = \underbrace{\Pi_{i,t}^e}_{\text{Exp. profit}} - \underbrace{\theta L_{i,t-1}}_{\text{Deb instalment}} - \underbrace{\tau^{\text{FIRM}} \max(0, \Pi_{i,t}^e)}_{\text{Corporate taxes}} - \underbrace{\theta^{\text{DIV}} (1 - \tau^{\text{FIRM}}) \max(0, \Pi_{i,t}^e)}_{\text{Dividend payout}}, \quad (5)$$

Return

Profits

Firm profits $\Pi_{i,t}$ are an accounting measure.

$$\begin{aligned}
 \Pi_{i,t} = & \underbrace{P_{i,t} Q_{i,t}}_{\text{Sales}} + \underbrace{P_{i,t} \Delta S_{i,t}}_{\text{Inventory change}} - \underbrace{\left(1 + \tau^{\text{SIF}}\right) w_{i,t} N_{i,t} \bar{P}_t^{\text{HH}}}_{\text{Labour costs}} \\
 & - \underbrace{\frac{1}{\beta_i} \bar{P}_{i,t} Y_{i,t}}_{\text{Material costs}} - \underbrace{\frac{\delta_i}{\kappa_i} P_{i,t}^{\text{CF}} Y_{i,t}}_{\text{Depreciation}} - \underbrace{\tau_i^Y P_{i,t} Y_{i,t} - \tau_i^K P_{i,t} Y_{i,t}}_{\text{Net taxes/subsidies on products/production}} \\
 & - \underbrace{r_t (L_{i,t-1} + \max(0, -D_{i,t-1}))}_{\text{Interest payments}} + \underbrace{\bar{r}_t \max(0, D_{i,t-1})}_{\text{Interest received}}
 \end{aligned} \tag{6}$$

r_t is the interest rate paid on outstanding loans.

Return

Cash Flow

Firm net cash flow is a measure how much liquidity is available for firms to use and is equal to changes in the deposit account:

$$\begin{aligned}
 \Delta D_{i,t} = & \underbrace{P_{i,t} Q_{i,t}}_{\text{Sales}} - \underbrace{\left(1 + \tau^{\text{SIF}}\right) w_{i,t} N_{i,t} \bar{P}_t^{\text{HH}}}_{\text{Labor costs}} - \underbrace{\bar{P}_{i,t} \Delta M_{i,t}}_{\text{Material costs}} \\
 & - \underbrace{\tau_i^Y P_{i,t} Y_{i,t} - \tau_i^K P_{i,t} Y_{i,t}}_{\text{Net taxes/subsidies on products and production}} - \underbrace{\tau^{\text{FIRM}} \max(0, \Pi_{i,t})}_{\text{Corporate tax payments}} \\
 & - \underbrace{\theta^{\text{DIV}} \left(1 - \tau^{\text{FIRM}}\right) \max(0, \Pi_{i,t})}_{\text{Dividend payments}} - \underbrace{r (L_{i,t-1} + \max(0, -D_{i,t-1}))}_{\text{Interest payments}} \quad (7) \\
 & + \underbrace{\bar{r}_t \max(0, D_{i,t-1})}_{\text{Interest received}} - \underbrace{P_{i,t}^{\text{CF}} l_{i,t}}_{\text{Investment costs}} + \underbrace{\Delta L_{i,t}}_{\text{New credit}} - \underbrace{\theta L_{i,t-1}}_{\text{Debt instalment}} .
 \end{aligned}$$

[Return](#)

More Accounting and Insolvency

- Firm deposits are equal to deposits plus net cash flow:

$$D_{i,t} = D_{i,t-1} + \Delta D_{i,t} \quad (8)$$

- Debt follows the following law of motion:

$$L_{i,t} = (1 - \theta)L_{i,t-1} + \Delta L_{i,t} \quad (9)$$

- Firm equity $E_{i,t}$ is computed as a balancing item

$$E_{i,t} = D_{i,t} + \sum_g a_{sg} \bar{P}_{g,t} M_{i,t} + P_{i,t} S_{i,t} + \bar{P}_t^{\text{CF}} K_{i,t} - L_{i,t} \quad \forall i \in I_s \quad (10)$$

- When a firm has negative balance on their deposit account (cash flow insolvent, i.e. $D_{i,t} < 0$) and has negative equity value (balance sheet insolvent, i.e. $E_{i,t} < 0$), they become insolvent.

Return

Government Overview

- Government purchases consumption on goods market
- Government levies taxes and takes social security contributions
- Government pays out social transfers

[Return](#)

Government Revenue

Taxes, social contributions and other transfers from all sectors form the revenue of the government:

$$\begin{aligned}
 Y_t^G = & \underbrace{\left(\tau^{\text{SIF}} + \tau^{\text{SIW}} \right) \bar{P}_t^{\text{HH}} \sum_{h \in H_t^{\text{E}}} w_{h,t}}_{\text{Social security contributions}} + \underbrace{\tau^{\text{INC}} \left(1 - \tau^{\text{SIW}} \right) \bar{P}_t^{\text{HH}} \sum_{h \in H_t^{\text{E}}} w_{h,t}}_{\text{Labor income taxes}} \\
 & + \underbrace{\tau_h^{\text{VAT}} \sum_h C_{h,t}}_{\text{Value added taxes}} + \underbrace{\tau^{\text{INC}} \left(1 - \tau^{\text{FIRM}} \right) \theta^{\text{DIV}} \left(\sum_i \max(0, \Pi_{i,t}) + \max(0, \Pi_{k,t}) \right)}_{\text{Capital income taxes}} \\
 & + \underbrace{\tau^{\text{FIRM}} \left(\sum_i \max(0, \Pi_{i,t}) + \max(0, \Pi_{k,t}) \right)}_{\text{Corporate income taxes}} + \underbrace{\tau^{\text{CF}} \sum_h I_{h,t}}_{\text{Taxes on capital formation}} \\
 & + \underbrace{\sum_i \tau_i^{\text{Y}} P_{i,t} Y_{i,t}}_{\text{Net taxes/subsidies on products}} + \underbrace{\sum_i \tau_i^{\text{K}} P_{i,t} Y_{i,t}}_{\text{Net taxes/subsidies on production}} + \underbrace{\tau_l^{\text{EXPORT}} \sum_l C_{l,t}}_{\text{Export taxes}}.
 \end{aligned}$$

(11)

Government Deficit and Debt

The government deficit (or surplus) is an accounting identity consisting of expenditures minus revenues:

$$\begin{aligned}
 \Pi_t^G = & \underbrace{\sum_{h \in H^{\text{inact}}} \bar{P}_t^{\text{HH}} sb_t^{\text{inact}} + \sum_{h \in H_t^{\text{U}}} \bar{P}_t^{\text{HH}} \theta^{\text{UB}} w_{h,t} + \sum_h \bar{P}_t^{\text{HH}} sb_t^{\text{other}}}_{\text{Social benefits and transfers}} \\
 & + \underbrace{\sum_j C_{Pj,t}}_{\text{Government consumption}} + \underbrace{r^G L_{t-1}^G}_{\text{Interest payments}} - \underbrace{Y_t^G}_{\text{Government revenues}}.
 \end{aligned} \tag{12}$$

The government debt is previous period government debt plus the current deficit:

$$L_t^G = L_{t-1}^G + \Pi_t^G. \tag{13}$$

Return

Banking Sector

- The bank provides loans to firms according to a risk assessment and a minimum capital requirement.
- The bank has to form expectations to be able to do so.
- The bank also holds deposits or provides credit to households.
- The bank receives interest payments and also needs to pay interest payments.

[Return](#)

Bank loan provision

The bank has to adhere to a minimum capital requirement coefficient $0 < \zeta < 1$:

$$\frac{E_{k,t}^e}{\sum_{i=1}^I (L_{i,t}^e + \Delta L_{i,t})} = \frac{E_{k,t-1}}{\sum_{i=1}^I ((1-\theta)L_{i,t-1} + \Delta L_{i,t})} \geq \zeta, \quad (14)$$

Due to fundamental uncertainty, the bank has to form expectations on the value of firm i 's capital stock ($K_i^e(t)$):

$$\frac{L_{i,t}^e + \Delta L_{i,t}}{K_{i,t}^e} = \frac{\overbrace{(1-\theta)L_{i,t-1} + \Delta L_{i,t}}^{=L_{i,t}^e}}{\underbrace{\bar{P}_{t-1}^{\text{CF}} (1 + \pi_t^e) K_{i,t-1}}_{=K_{i,t}^e}} \leq \zeta^{\text{LTV}}. \quad (15)$$

Return

Bank Profits

The difference between interest revenues from loans provisions and costs due to interest payments on deposits held with the bank by firms and households are the bank's profits.

$$\Pi_{k,t} = r_t \underbrace{\left(\sum_{i=1}^I L_{i,t-1} + \max(0, -D_{i,t-1}) + r_t \sum_{h=1}^H \max(0, -D_{h,t-1}) + \bar{r}_t \max(0, D_{k,t-1}) \right)}_{\text{Interest received}} - \underbrace{\bar{r}_t \sum_{i=1}^I \max(0, D_{i,t-1}) - \bar{r}_t \sum_{h=1}^H \max(0, D_{h,t-1}) - \bar{r}_t \max(0, -D_{k,t-1})}_{\text{Interest payments}} \quad (16)$$

The bank pays policy rate $\bar{r}_t$ on deposits. The effective interest rate r_t faced for bank credit or overdrafting is determined by the policy rate $\bar{r}_t$ and a fixed markup μ :

$$r_t = \bar{r}_t + \mu \quad (17)$$

Return

Bank Accounting

Bank equity is an accounting identity which is determined by bank profits or losses and is given by

$$\begin{aligned}
 E_{k,t} = & E_{k,t-1} + \Pi_{k,t} - \underbrace{\theta^{\text{DIV}} \left(1 - \tau^{\text{FIRM}} \right) \max(0, \Pi_{k,t})}_{\text{Dividend payments}} \\
 & - \underbrace{\tau^{\text{FIRM}} \max(0, \Pi_{k,t})}_{\text{Corporate taxes}} - \underbrace{\sum_{i \in I'} \left(L_{i,t} - D_{i,t} - \zeta^b \bar{P}_i^{\text{CF}}(t) K_{i,t} \right)}_{\text{Write-off of bad debt}}, \quad (18)
 \end{aligned}$$

Return

Central Bank

- Central bank sets policy rate according to Taylor rule.
- The central bank receives revenues from interest payments on government debt.
- The central bank receives or pays interest depending on the net position of the advances or reserves to the banking system.
- The central bank balance sheet determines whether the economy is a net creditor or net debtor to the rest of the world.

[Return](#)

Interest Rate Setting

A generalized Taylor rule (Taylor, 1993) determines the policy rate. We use the same rule as Blattner and Margaritov (2010), which includes a "growth" rule specification:

$$\bar{r}_t = \rho \bar{r}_{t-1} + (1 - \rho) \left(r^* + \pi^* + \xi^\pi \left(\pi_t^{\text{EA}} - \pi^* \right) + \xi^\gamma \gamma_t^{\text{EA}} \right) \quad (19)$$

where ρ is a measure for gradual adjustment of the policy rate, r^* is the real equilibrium interest rate, π^* is the inflation target of the central bank, ξ^π is the weight the central bank puts on inflation targeting and ξ^γ the weight placed on economic growth, respectively.

[Return](#)

Central Bank Accounting

The difference between revenues from interest payments on government debt, as well as revenues or costs is defined as the central bank's profits.

$$\Pi_t^{\text{CB}} = r^G L_{t-1}^G - \bar{r}_t D_{k,t-1}. \quad (20)$$

The central bank's equity E_t^{CB} in an accounting identity dependent on its profits or losses and its past equity.

$$E_t^{\text{CB}} = E_{t-1}^{\text{CB}} + \Pi_t^{\text{CB}}. \quad (21)$$

[Return](#)

Central Bank Accounting

The following law of motion determines the net creditor/debtor position of the national economy to the rest of the world (D_t^{Row}):

$$D_t^{\text{Row}} = D_{t-1}^{\text{Row}} - \underbrace{\left(1 + \tau^{\text{EXPORT}}\right) \sum_l C_{l,t}}_{\text{Exports}} + \underbrace{\sum_m P_{m,t} Q_{m,t}}_{\text{Imports}}. \quad (22)$$

This connects the change in the amount of deposits in the banking system to the government deficit (surplus) and to the balance of trade:

$$\begin{aligned} E_t^{\text{CB}} + D_t^{\text{Row}} &= L^G(t) - D_{k,t} \\ &= L_t^G - \sum_{i=1}^I D_{i,t} - \sum_{h=1}^H D_{h,t} - E_{k,t} + \sum_{i=1}^I L_{i,t}. \end{aligned} \quad (23)$$

Return

Rest of the World

- Supply of imports and demand of exports are determined by an AR(1) processes.
- Demand for imports and supply of exports are endogenous.
- Realized imports and exports are determined by the search-and-matching mechanism.

Extensive description

Imports

- Total supply of Imports Y_t^I in real terms follows an AR(1) rule.
- Thus the m -th, ($m \in \{1, \dots, S\}$), foreign firm representing an industry s imports the principal product g : $Y_{m,t} = c_{g=s}^I Y_t^I$.
- Import price is assumed to develop similarly as the average sectoral domestic adjusted for expected inflation: $P_{m,t} = \bar{P}_{g,t-1}(1 + \pi_t^e)$.
- Sales of imports are equal to the realized demand which is an outcome of the search-and-matching process on the goods markets:
 $Q_{m,t} = \min(Y_{m,t}, Q_{m,t}^d)$.

[Return](#)

Exports

- Total demand for export is C_t^E is assumed to be exogenous and follows an AR(1) rule.
- The total demand for exports is uniformly distributed to L foreign consumers and attributed to goods g :

$$C_{l,t}^d = \frac{C_t^E \sum_g c_{g,t}^E \bar{P}_{g,t-1} (1 + \pi_t^e)}{L}, \quad (24)$$

- The demand for exported goods by the l -th foreign consumer to purchase the g -th good is given by

$$C_{lg,t}^d = \frac{c_g^E \bar{P}_{g,t-1}}{\sum_g c_g^E \bar{P}_{g,t-1}} C_{l,t}^d. \quad (25)$$

- Realized consumption by foreign consumers is again determined by the search-and-matching process on goods markets.

Extensive description